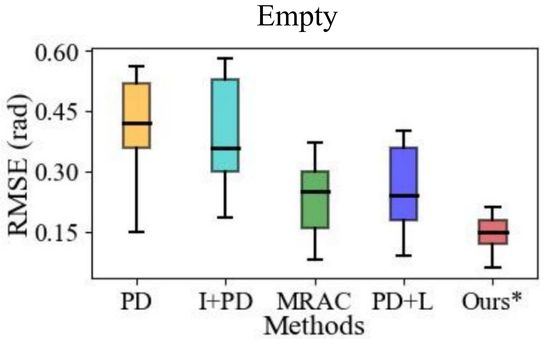
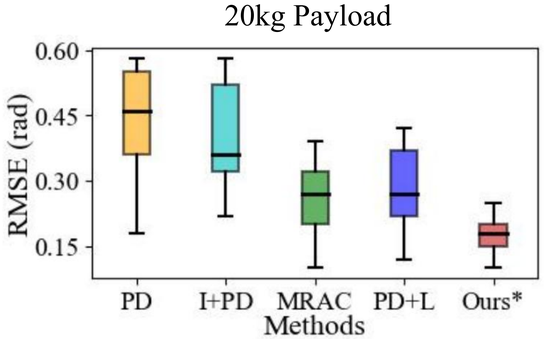
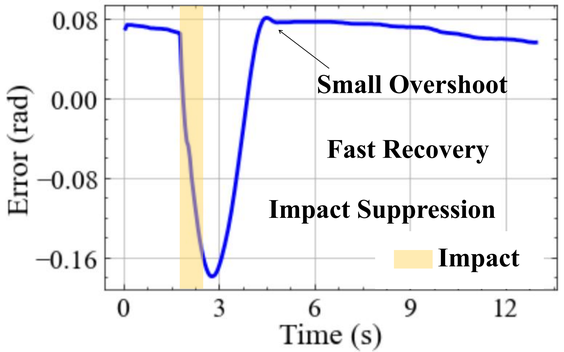
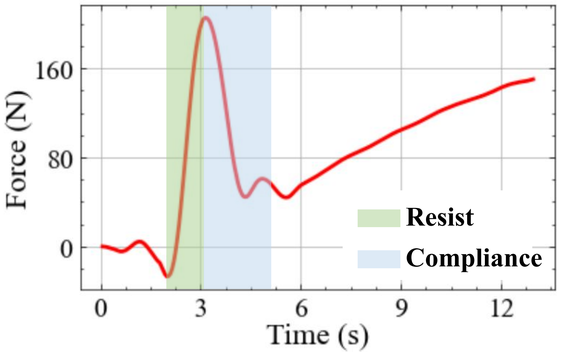

(a)

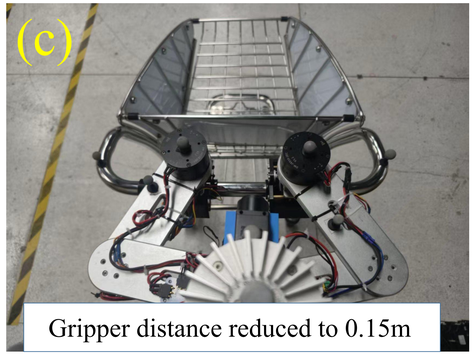

(b)

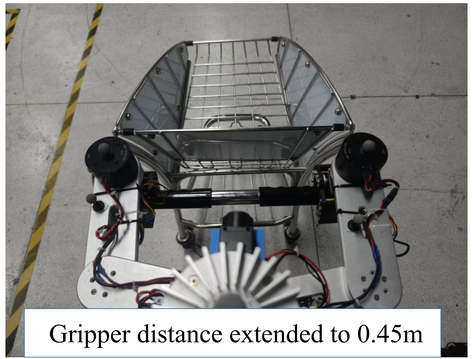

Empty
RMSE (rad)
0.60
0.45
0.30
0.15
PD
I+PD
MRAC
PD+L
Ours*
Methods

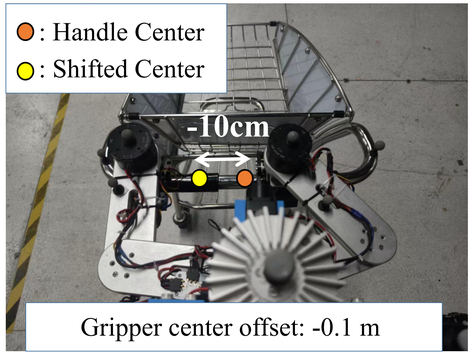

20kg Payload
RMSE (rad)
0.60
0.45
0.30
0.15
PD
I+PD
MRAC
PD+L
Ours*
Methods

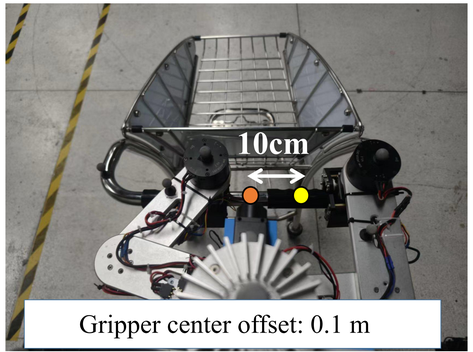

Error (rad)
0.08
0.00
−0.08
−0.16
Small Overshoot
Fast Recovery
Impact Suppression
Impact
0
3
6
9
12
Time (s)


Force (N)
160
80
0
Resist
Compliance
0
3
6
9
12
Time (s)


(c)
Gripper distance reduced to 0.15m


Gripper distance extended to 0.45m


: Handle Center
: Shifted Center
-10cm
Gripper center offset: -0.1 m


10cm
Gripper center offset: 0.1 m